

# Brian Cahill

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## Professional Summary

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Engineer with end-to-end deployment experience across large-scale robotic infrastructure, owning technical execution and cross-functional coordination for automated fulfillment centers running up to 8,000 autonomous mobile robots. Uses data driven decision making and analysis to surface and correct operational inefficiencies at scale, translating findings into direct cost impact. Now seeking a customer-facing role where technical depth and effective communication translate to business outcomes.

## Work Experience

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**Amazon Robotics** North Reading, MA

April 2024 – Current

**Solutions Design Engineer | Lead Solutions Deployment Engineer, Robotic Storage Technology**

- Designed robotic warehouse solutions for automated storage retrieval systems running up to 8,000 autonomous mobile robots across 750k+ sq. ft facilities converting building outbound specifications into operational solutions
- Led end-to-end deployment for all North American new building and retrofit solutions, managing 6 concurrent deployments with accountability for equipment install, robotic path design, and deployment documentation
- Discovered and resolved a 7% surplus of an outdated storage bin at one site costing ~14,000 cu ft and \$1.1 million in lost storage
- Created an internal knowledge base converting design documentation and drawing structure parameters into a LISP chat agent that intakes a design issue or manual task and outputs LISP commands that can be seamlessly run in AutoCAD. Two week beta test with four designers has decreased CAD time by 40% equating to ~200 hours per designer annually
- Developed Python automation tools eliminating 20+ hours of manual work per designer per deployment including a wireless Access Point placement script (reduced 8+ hour task to <5 minutes)
- Delivered two critical site designs in <2 business days (typical timeline: 2 weeks), saving \$192k in remobilization costs alone and preventing project delays and more substantial losses
- Built comprehensive Deployment Wiki establishing standardized processes and automating control checklists saving team members 60+ hours and on site team ~\$55k in change orders per deployment
- Organized a weekly cross functional sync with hardware and deployment customers to answer all questions and resolve high severity issues

**AcousticaBio** Cambridge, MA

January 2023 – August 2023

**Automation Engineering Co-op**

- Built a Python (OpenCV) & ImageJ analysis pipeline to process video and image data from microparticle experiments, generating automated statistics and distribution plots to quantify printer performance
- Coordinated and executed microparticle concentration and size experiments, providing quantitative data to guide hardware iteration decisions
- Designed, prototyped, and validated mechanical components of an acoustophoretic bioprinter capable of generating a 250g force

**Raytheon Missiles & Defense** Andover, MA

July 2021 – January 2022

**Operations Engineering Co-op**

- Led AutoCAD documentation effort to reorganize 100,000+ sq. ft. of factory floor space at the Andover facility, coordinating across multiple team leads and identifying previously unrecognized layout optimization opportunities
- Established AutoCAD floor plan documentation standards for the Andover Industrial Engineering team, creating a repeatable framework for ongoing facility mapping
- Built a VBA QR scanner log enabling floor operators to track Work-in-Progress in real time, reducing material loss and improving production traceability

## Education

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**Northeastern University** Boston, MA

December 2023

Bachelor of Science in Mechanical Engineering, Minor in Data Science | GPA: 3.80/4.00

Awards: Magna Cum Laude, Dean's List (all semesters), 1st place in Mechanical Engineering Capstone track

**Senior Capstone Project Metal Powder Aerosolizer:** Created a novel method to aerosolize fine powders using vibrating ERM motors and a pressure differential system for cold spray 3D printing applications

## Skills & Certifications

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**Programming & Automation:** Python (Scikit-learn, NumPy, Pandas, etc.), AutoLISP, SQL, VBA, MATLAB, C++, ImageJ

**System & Design:** Robotic Path Planning, AutoCAD, SolidWorks (CSWA), ASM, FEA, Arduino

**Certifications:** DoD Secret Clearance (2021), SolidWorks CSWA, Bloomberg

## Interests

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Enthusiastic about golf, hockey, football, and baseball; enjoy hiking, personal fitness, and tv shows